Group1

2026-04-16

Introduction

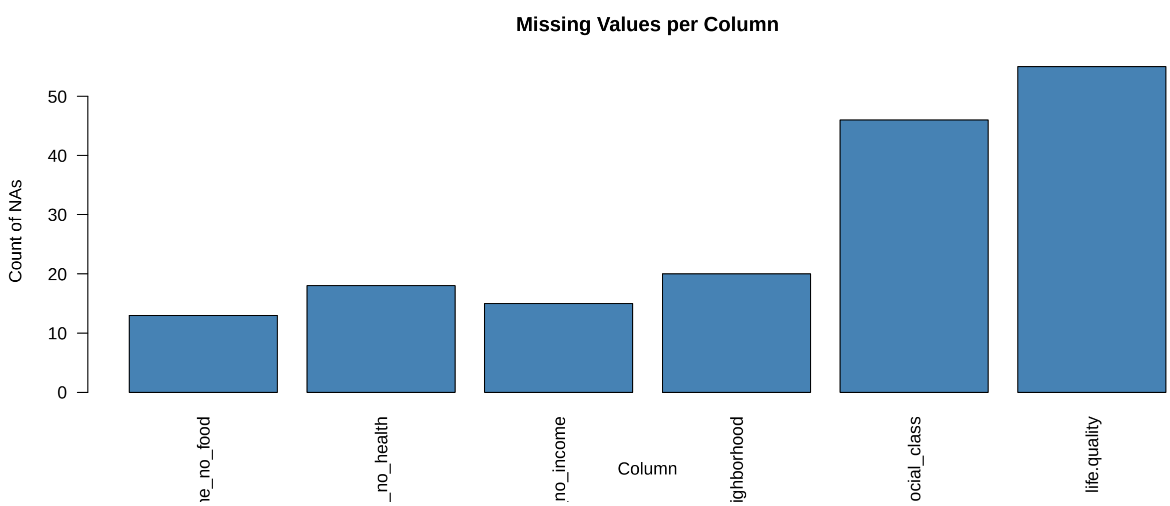
Literature Review

Theory

Hypothesis

Exploratory Data Analysis

Missing Data Plot



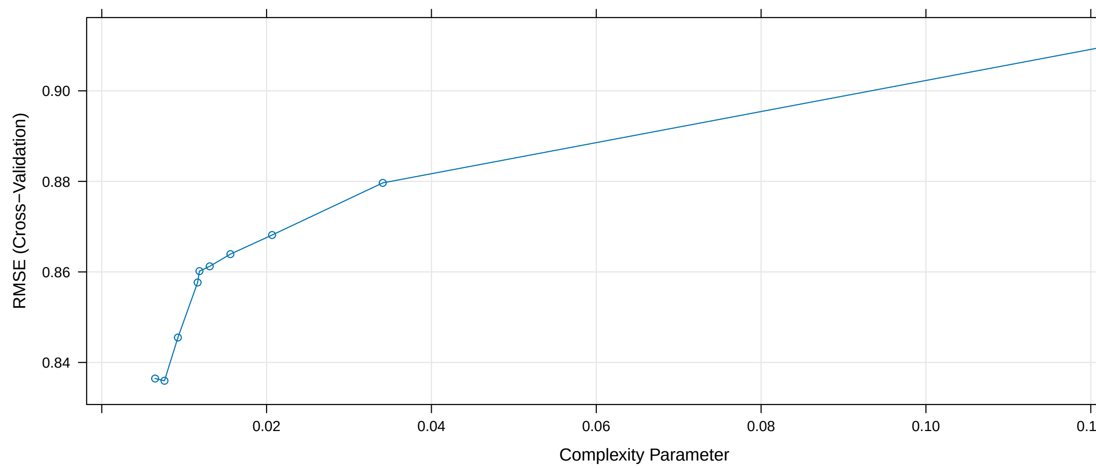
Data Cleaning

Methodology

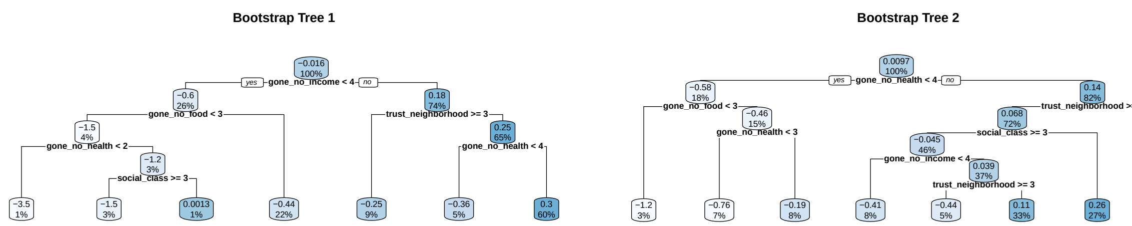
Assumptions

Testing and Validation

Unpruned and Pruned trees comparison

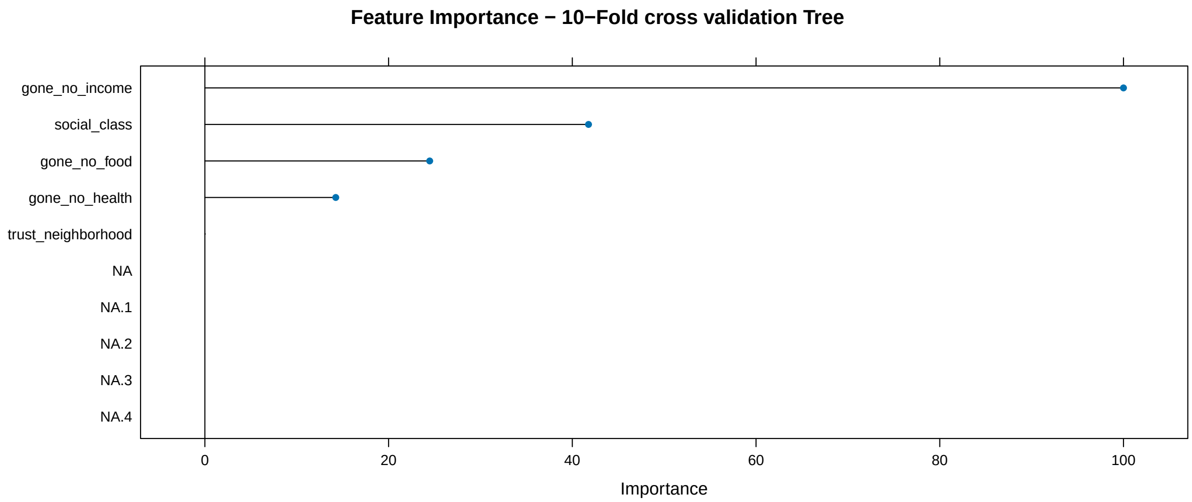


K-Folds cross validation



Bagging validation:
Where we try to observe the different decision trees formed by using bootstrap samples of the dataset provided for our interested features. ## Result Analysis

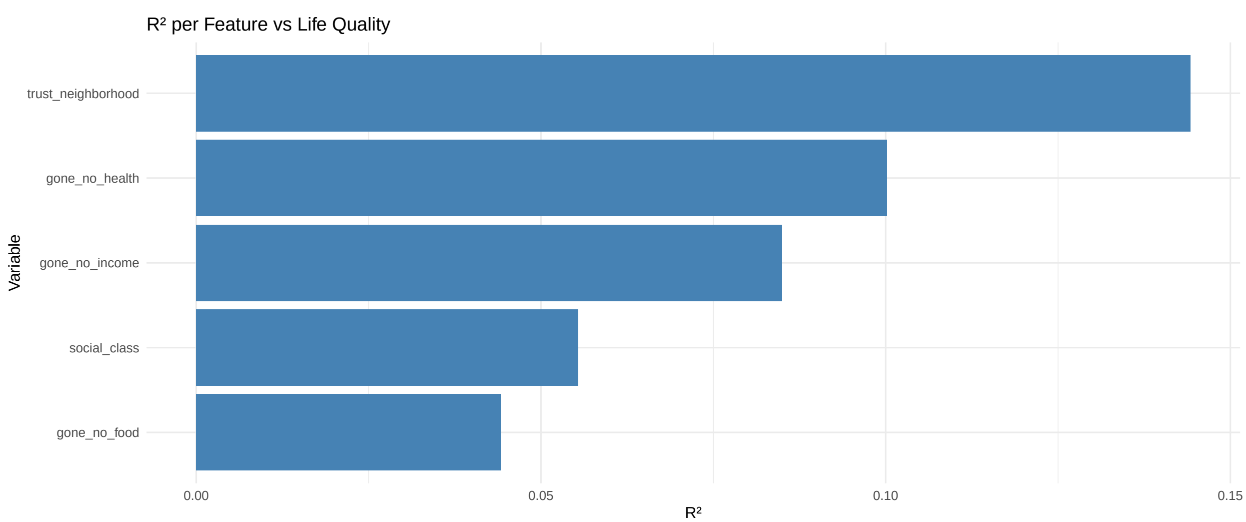
Feature Importance observed through K-folds validation where the number of folds = 10



The goodness of fit (R^2) for each model and the root mean squared error for each model

```
##            Model   RMSE      R2
## 4        Bagging 0.8907  0.1161
## 2    Pruned Tree 0.9231  0.0506
## 3        CV Tree 0.9269  0.0428
## 1  Unpruned Tree 1.0222 -0.1642
```

Plotting goodness of fit for each feature R^2 used in the decision tree.

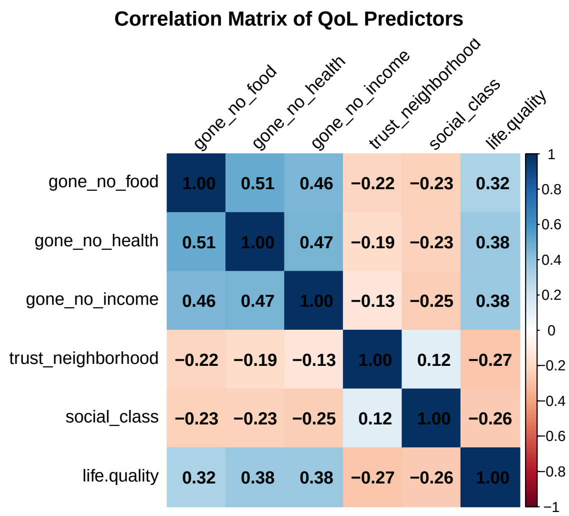


Testing Decision Tree Validity and Reliability

Correlation matrix

```
##                      gone_no_food gone_no_health gone_no_income
## gone_no_food            1.0000000      0.5062064      0.4554664
```

```
## gone_no_health        0.5062064     1.0000000     0.4688423
## gone_no_income        0.4554664     0.4688423     1.0000000
## trust_neighborhood   -0.2173007    -0.1879949    -0.1325449
## social_class         -0.2343073    -0.2252231    -0.2453556
## life.quality          0.3188351     0.3757251     0.3769237
##                    trust_neighborhood social_class life.quality
## gone_no_food               -0.2173007   -0.2343073    0.3188351
## gone_no_health             -0.1879949   -0.2252231    0.3757251
## gone_no_income             -0.1325449   -0.2453556    0.3769237
## trust_neighborhood          1.0000000    0.1213683   -0.2721205
## social_class                0.1213683    1.0000000   -0.2554861
## life.quality               -0.2721205   -0.2554861    1.0000000
```

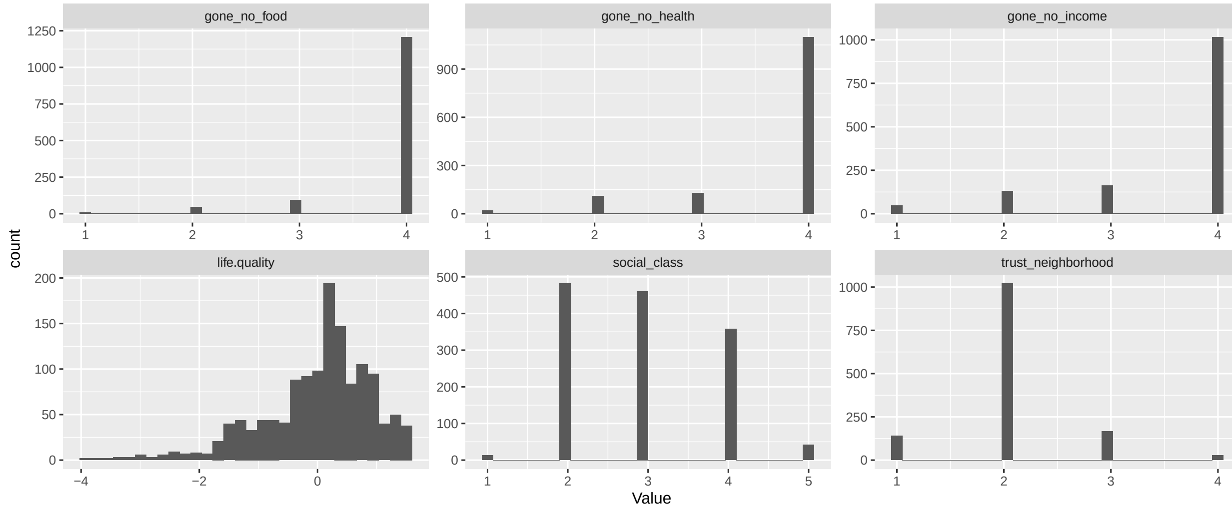


Observations - Positive correlation between health, income and food although the relationship itself is quite weak (score of 0.29 with food and score of 0.36 with both health and food). - All three variables have positive correlations score with each other, indicating that these factors affect each other. - Very weak positive relationship between trust in neighbourhoods and social class.

Skewness Checks

```
##                    skewness
## gone_no_food         -3.447
## gone_no_health       -2.234
## gone_no_income       -1.835
## life.quality         -1.034
## trust_neighborhood    0.747
## social_class          0.281
```

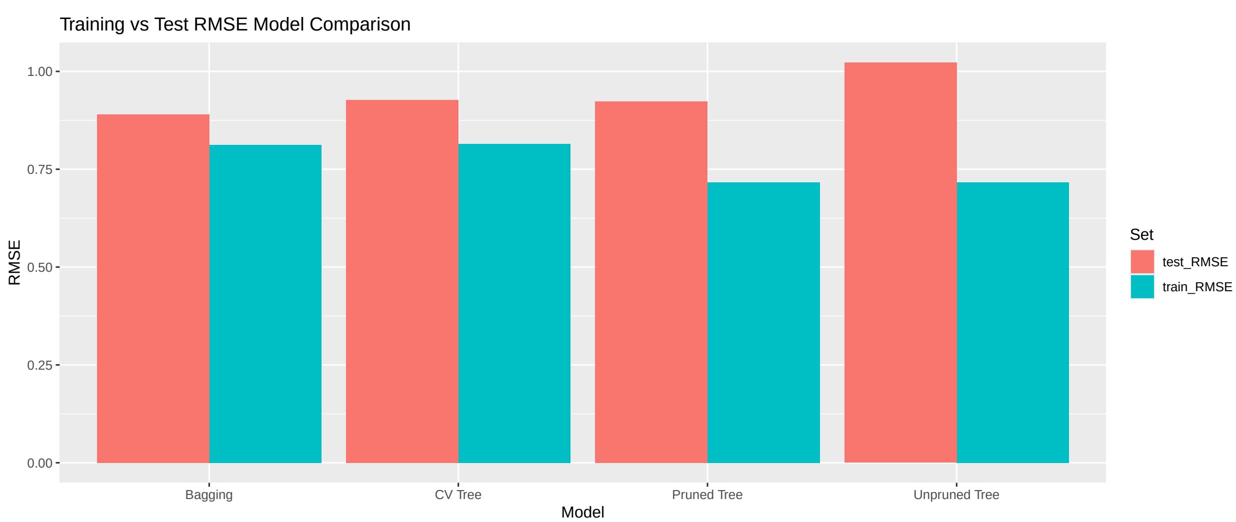
```r
# Pivot skew measurement data frame from wide to long
train[, c(predictors, "life.quality")] |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Value") |>
  ggplot(aes(x=Value)) +
  geom_histogram(bins = 30) + facet_wrap(~Variable, scales="free")
```



Discussion - For variables `gone_no_food`, `gone_no_health`, `gone_no_income`, majority of survey responses sit at highest possible score. This also confirms left-skew distribution for `life.quality`. - Very few respondents indicated poor quality of life factors (not enough food, bad health and low income). - Indicates that ceiling effect might be occurring, a measurement limitation that is occurring in the survey due to a low upper limit. - Graph distribution of `life.quality` values show a left skew, these match skewness values shown above. - `social_class` more balanced, values concentrated on response values 2-4

Detecting model for overfitting

```
##              Model train_RMSE test_RMSE RMSE_gap
## 4          Bagging     0.8123    0.8907   0.0784
## 3          CV Tree     0.8153    0.9269   0.1116
## 2      Pruned Tree     0.7171    0.9231   0.2060
## 1 Unpruned Tree     0.7171    1.0222   0.3051
```



Discussion

- Tree trained with cross-validation saw the smallest gap in RMSE, indicating this model strongly generalises on unseen data.

- Unpruned Tree model had the largest gap between train and test RMSE, indicating this model is more prone to overfitting on unseen data.
- Both pruned and unpruned tree saw the same RMSE on training data, however the pruned tree was less error-prone on the test data, showing the effectiveness of pruning the decision tree.

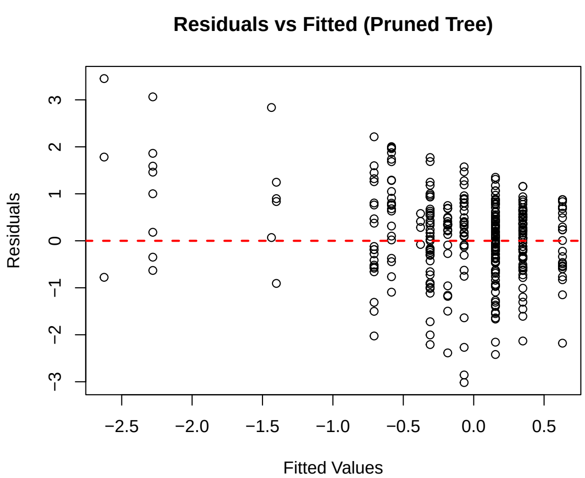
Pre-pruning Exploration

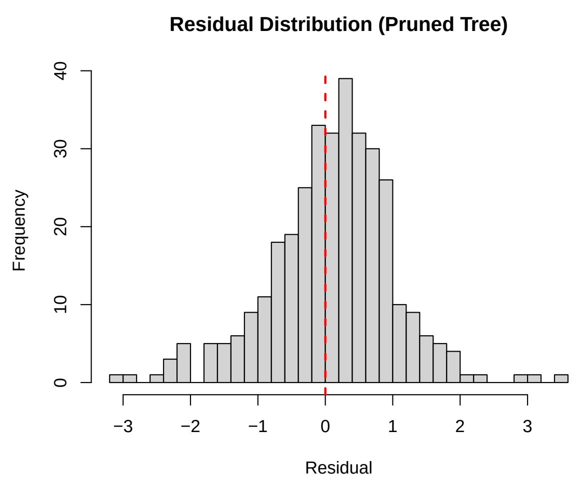
```
##   MaxDepth MinSplit MinBucket   RMSE     R2 N_leaves
## 4       15       30        15 0.9218 0.0531        7
## 2        5       10         5 0.9272 0.0421        8
## 3       10       20        10 0.9272 0.0421        8
## 1        3       10         5 0.9315 0.0333        7
```

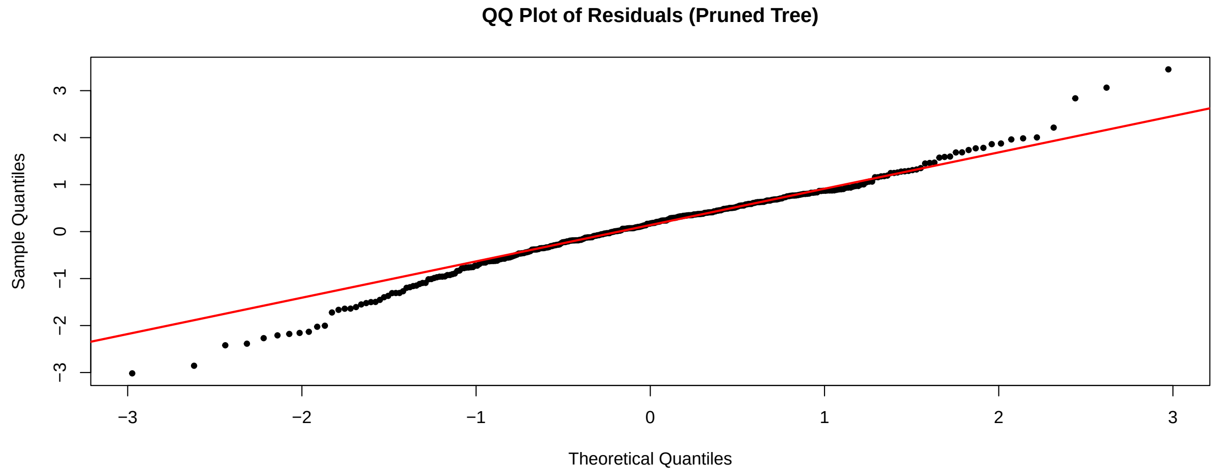
Discussion

- Pre-pruning reveals that the decision tree naturally stops growth, produces 7-8 leaves regardless of what is specified as the maximum depth of the tree.
- Consistently low R^2 value, ranging from ~0.03 to ~0.05. Ties back into the point raised about ceiling effects
- All variants reveal similar RMSE values

Residual Analysis








```
## Residual Summary:
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## -3.01721 -0.38205  0.17301  0.09829  0.66114  3.45295
```

```
##
## Mean Residual: 0.0983
```

```
## SD of Residuals: 0.9192
```

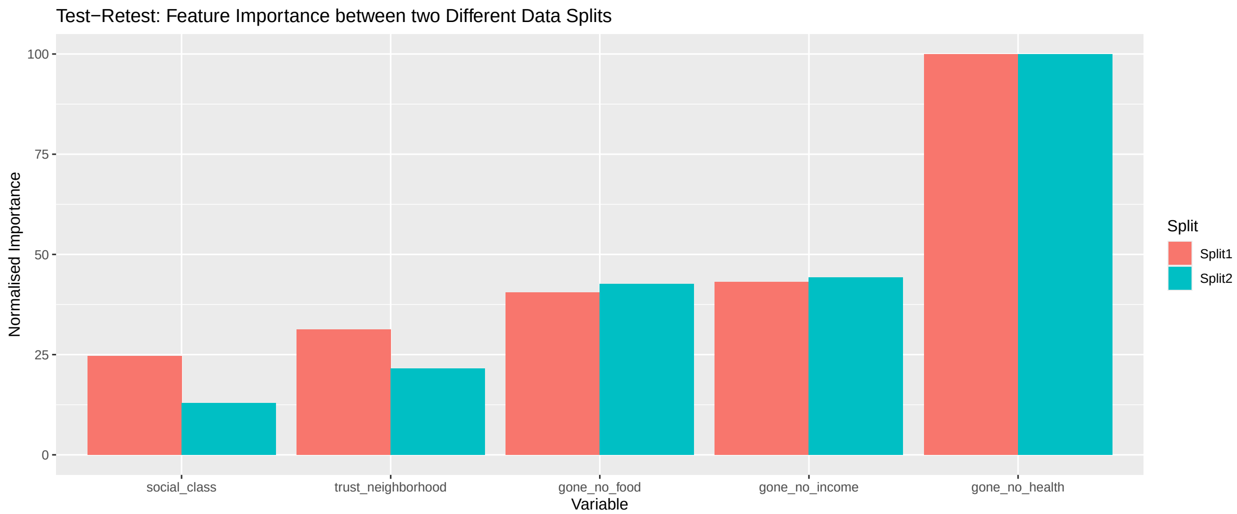
Discussion

- Residual vs fitted plot displays no systemic bias, points follow shape of vertical columns but this is expected from survey data and decision tree nature (fixed leaf nodes).
- Residual distribution shows a very close bell-shape, mostly centered near zero.
- QQ Plot confirms near-normality of data, points follow diagonal very closely in the middle and minimal deviation at both tails of the graph
- Mean residual of 0.098 shows slight tendency to under-predict QoL.
- Plots show that model is overall valid, especially with how distribution of residuals how the QQ Plot is shaped.

Test-Retest Reliability

```
##                                Variable Split1 Split2
## gone_no_health         gone_no_health 100.00 100.00
## gone_no_income         gone_no_income  43.19  44.26
## gone_no_food             gone_no_food  40.53  42.59
## trust_neighborhood trust_neighborhood  31.29  21.53
## social_class             social_class  24.63  12.94
```

```
## Feature Importance Rank Correlation (Spearman): 1.000
```



Discussion - Strong consistency in rankings between the two splits - `gone_no_health` the most dominant predictor (scoring 100 in both splits) - `gone_no_income` and `gone_no_food` demonstrated strong stability in both splits, minimal score deviation from first split to second split (~1 point difference for `gone_no_income` and ~2 point difference for `gone_no_food`) - `trust_neighbourhood` and `social_class` showed greater variation in feature importance score, still retained same rank in both splits.

Limitations

References